

Program : Diploma in Renewable Energy	
Course Code : 6292B	Course Title: Electrical Energy Conservation
Semester : 6	Credits: 4
Course Category: Open Elective course	
Periods per week: 4 (L:4 T:0 P:0)	Periods per semester: 60

Course Objectives:

- To understand the need for energy conservation.
- To understand electrical energy conservation techniques.

Course Prerequisites:

Topic	Course Code	Course name	Semester
Work, Power and Energy		Applied Physics	1

Course Outcomes:

On completion of the course, the student will be able to:

COn	Description	Duration (Hours)	Cognitive Level
CO1	Explain the importance of energy conservation and management.	14	Understanding
CO2	Explain various renewable energy sources and its importance.	15	Understanding
CO3	Summarize energy conservation techniques in electrical installations and industries	13	Understanding
CO4	Explain various practices for energy conservation	16	Understanding
	Series Test	2	

CO – PO Mapping

Course Outcomes	PO1	PO2	PO3	PO4	PO5	PO6	PO7
CO1	2				3		
CO2	2				3		
CO3	2				3		
CO4	2				3		

3-Strongly mapped, 2-Moderately mapped, 1-Weakly mapped

Course Outline

Module Outcomes	Description	Duration (Hours)	Cognitive Level
CO1	Explain the importance of energy conservation and management.		
M1.01	Outline the current energy scenario in India.	3	Understanding
M1.02	Outline the importance and principles of energy conservation	3	Understanding
M1.03	Explain various energy management techniques.	5	Understanding
M1.04	Explain various elements of energy efficiency.	3	Understanding
Contents: Energy Scenario: current global energy scenario-coal, oil and natural gas reserve-statistical data- energy consumption in various sectors in india. (review in industrial, agriculture and residential sectors)-long term energy scenario for india-environmental impacts due to the use of conventional energy sources- solutions for environmental issues. Energy conservation- definition-need- importance -principles. Energy management- definition-objectives-functions- importance of energy management-energy management techniques - self knowledge and awareness-reengineering and evaluation-technology upgradations. Energy efficiency- concept- benefits- elements of energy efficiency-efficient energy usage-energy conversion efficiency-energy conservation			
CO2	Explain various renewable energy sources and its importance.		

M2.01	Classify various energy sources with examples	2	Understanding
M2.02	Illustrate various methods of solar energy conversion	4	Understanding
M2.03	Explain the wind and ocean energy conversion systems	5	Understanding
M2.04	Illustrate the energy conversion from geothermal and bio resources	4	Understanding
	Series Test – I	1	
Contents: Energy sources -classification- examples -conventional -non conventional-need of renewable energy -advantages- disadvantages Solar energy -basic principle -advantages- disadvantages- applications.-solar photovoltaic cell- definitions -solar radiation- solar irradiation- solar insolation. Wind energy -basic principle -advantages- disadvantages- applications.- components of a typical wind energy system-Block diagram. Ocean energy -basic principle -tidal energy-wave energy--advantages- disadvantages Geothermal Energy -basic principle- resources-advantages- disadvantages Bio gas -definition-list the constituents -applications -biogas plants-types -construction -working-dome type - drum type. Biomass - definition- various resources			
CO3	Summarize energy conservation techniques in electrical installations and industries		
M3.01	Explain energy conservation techniques in lighting systems.	4	Understanding
M3.02	Outline energy conservation opportunities in induction motor	5	Understanding
M3.03	Explain various energy conservation equipments for electric motors	3	Understanding
M3.04	Outline energy conservation opportunities in pumps	1	Understanding

Contents :

Energy conservation techniques in lighting systems-General energy saving opportunities in lighting -Replacing Lamp sources-Using energy efficient luminaries.(lighting techniques – Natural , CFL, LED lighting sources and fittings.)

Energy Conservation techniques in fans-Electronic regulators

Energy saving opportunities in electric motors:Induction motors- motor efficiency-Using energy conservation equipments like, Soft starters, Power factor controller, Automatic, star-delta converter,(basic concept only),Advantages of soft start.Periodic maintenance: Good maintenance practices

Energy efficient motors: Technical aspects of energy efficient motors.

Energy conservation opportunities in pumping systems(Listing only)

CO4	Explain various practices for energy conservation		
M4.01	Explain the various energy conservation activities in buildings	4	Understanding
M4.02	Summarize various energy conservation opportunities in HVAC system.	4	Understanding
M4.03	Choose suitable tariff system to reduce electricity bill	4	Understanding
M4.04	Interpret star labelling, needs and its benefits.	4	Understanding
	Series Test – II	1	

Contents:

Energy conservation in buildings- energy saving tips- domestic sector- commercial sector- energy conservation activities in buildings

HVAC system-components- energy conservation opportunities in HVAC system-thermal insulation and energy efficiency-thermal energy conservation opportunities in buildings automatic door closing- Air curtain,

Tariff: Types of tariff structure: Special tariffs; Time-off-day tariff, Peak-off-day tariff, Power factor tariff, Maximum Demand tariff, Load factor tariff. Application of tariff system to reduce energy bill

Star labeling-need - benefits- types of label- comparative label-endorsement label-criteria for selection of home appliances --BEE (Bureau of Energy Efficiency) star rating-list of appliances covered under Standards & Labeling program-

Text / Reference

T/R	Book Title/Author
T1	Electric Energy Generation, Utilisation and Conservation, Sivaganaraju, S, Pearson, New Delhi
T2	Art and Science of utilization of Electrical Energy, H. Partab, Dhanapat Rai and Sons, New Dehli
T3	Fundamentals of electrical system- Bureau of Energy Efficiency
T4	Guide Books No. 1 and 3 for National Certification Examination for Energy Managers and Energy Auditors, Bureau of Energy Efficiency (BEE), Bureau of Energy Efficiency (A Statutory body under Ministry of Power, Government of India) (Fourth Edition 2015)
R1	Renewable Energy Resources: Basic Principles and ApplicationsG. N. Tiwari, M. K. Ghosal
R2	Singh, Sanjeev; Rathore, Umesh, Energy Management, S K Kataria&Sons,New Delhi ISBN-13:

Online Resources

Sl.No	Website Link
1	<u>Guide Books for National Certification Examination</u> https://aipnpc.org/Guidebooks.aspx
2	<u>NPTEL :: Electrical Engineering - Energy Resources & Technology</u>
3	<u>NPTEL :: Electrical Engineering - Non-Conventional Energy Systems</u>